

Assignment - 05
Computer Network

10/12/23

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• Section-A ($5 \times 2 = 10$)

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(1) What is application layer?

Ans The application layer is the topmost layer in the OSI (Open system Interconnection) model and the TCP/IP stack. It is responsible for providing network services directly to end-users or applications. The primary goal of the application layer is to enable communication between software applications on different devices. Common protocols and services associated with the application layer include HTTP for web browsing, SMTP (Simple mail transfer Protocol) for email, FTP (File transfer protocol) for file transfer and DNS for domain name resolution. The application layer serves as the interface betⁿ the network and the S/W applications, enabling comm. and data exchange b/w different systems.

2. Define the term DNS?

Ans DNS stands for Domain Name system. DNS is a directory service that provides a mapping between the name of a host on the network and its numerical address. DNS is required for the functioning of the internet. DNS is a service that translates the domain name into IP addresses. This allows the users of networks to utilize user-friendly names when looking for other hosts instead of remembering the IP addresses.

Q. What is electronic mail?

Ans Electronic mail, commonly known as email, refers to the exchange of digital messages between people using electronic devices like computers, tablets and smartphones. Email has become a widely used method

of communication for personal, professional and business purpose. It allows individuals to send and receive message, document and other files over the Internet.

(4) What do you mean by HTTP?

Ans: HTTP stands for Hypertext Transfer Protocol. It is the foundation of data communication on the World Wide Web and is used for transferring hypertext, which can include text, images, videos, and other multimedia files. HTTP is an application layer protocol, part of the Internet Protocol suite, and is used for the communication between web browser and web servers.

(5) What is simple network management protocol?

→ Simple Network Management Protocol, or SNMP, is a protocol widely used for managing and monitoring network devices and systems in a network. SNMP is an application layer protocol that facilitates the exchange of management information between network devices. It is a part of the Internet Protocol Suite (TCP/IP) and is an essential component in network management systems. SNMP provides a standardized way for network administrators to monitor and manage network performance, troubleshoot issues, and configure devices remotely. It is widely used in a variety of networked environments, including enterprise networks, telecommunications networks and the Internet.

• SNMP has two components Manager and Agent.

- Section - B ($3 \times 7 = 21$)

1. Explain the merits and demerits of application layer?

Ans → The application layer of the OSI model and TCP/IP stack plays a crucial role in enabling communication between software applications.

- Merit (Advantages) of the Application Layer:

- User-Friendly Communication** - The application layer provides a user-friendly interface for communication.
- High-level Abstraction**: The application layer abstracts the complexities of lower layers, allowing developers to focus on designing and implementing applications without needing in-depth knowledge of network protocols.
- Diverse Services**: It supports a wide range of network services, including email, file transfer, and directory services, to various communication needs.
- Platform Independence**: Application developed at the application layer are generally platform-independent. This independence allows users to access services from different devices and operating systems.

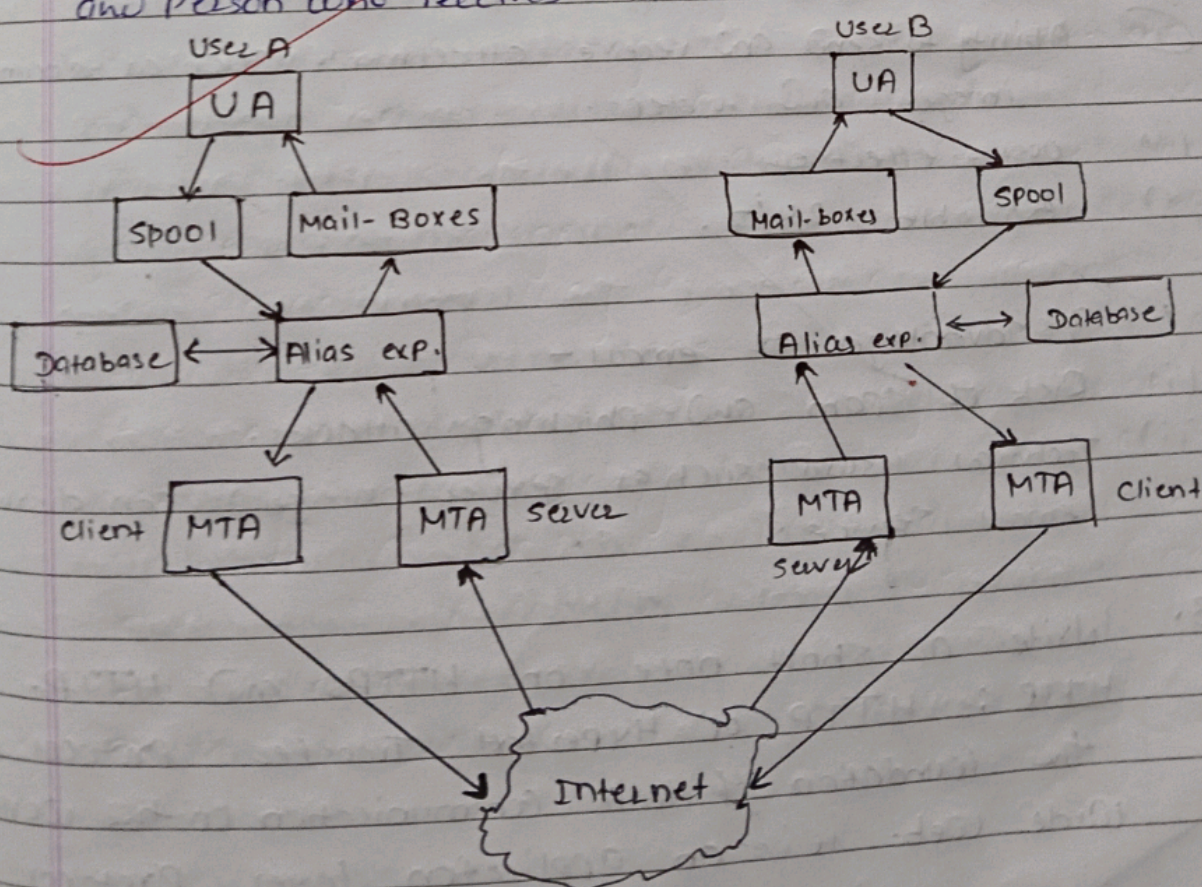
- Demerits (Disadvantages) of the Application Layer:

- Limited Control Over Network Resources**: Application layer protocols may have limited control over network resource and may not optimize data transfer efficiency compared to lower-layer protocols.
- Increased Overhead**: The app. layer introduces additional overhead due to the encapsulation and formatting of data for presentation to users. This overhead can impact network performance.

- (iii) **Security Concerns:** Security vulnerabilities at the application layer can pose risks to the entire system.
- (iv) **Dependence on lower layers:** While the application layer abstracts lower-layer complexities, it still depends on the proper functioning of lower-layer protocols and services for effective communication.

2. Describe the electronic mail with a diagram.

Ans. Electronic mail, commonly known as email, is a method of exchanging messages over the Internet. E-mail is one of most widely used services of Internet. This service allows an Internet user to send a message in formatted manner (mail) to the other Internet user in any part of world. Message in mail not only contain text, but it also contains images, audio and videos data. The person who is sending mail is called sender and person who receives mail is called recipient.



The basic components of an email system are: User Agent (UA), Message Transfer Agent (MTA), Mail Box, and Spool file.

• Service provided by E-mail system:

- Composition - The composition refers to the process that creates messages and sends them.
- Transfer - Transfer means the sending procedure of mail i.e. from the sender to the recipient.
- Reporting - Reporting refers to the confirmation for the delivery of mail. It helps the user to check whether their mail is delivered, lost or rejected.
- Displaying - It refers to presenting mail in a form that is understandable by the user.

• Advantages of email -

- (i) Fast communication with individuals or groups globally.
- (ii) Easy to store
- (iii) Ability to send and receive attachments such as documents, images, and videos.
- (iv) Cost-effective
- (v) Available 24/7.

• Disadvantages of email -

- (i) Risk of spam and phishing attacks.
- (ii) Technical issues, such as server outages, can disrupt email service.

✓ 3. Write a short note on HTTPS and HTTP.

✓ Ans HTTP: HTTP or Hypertext Transfer Protocol, is the foundation of data communication on the World Wide Web. It is an application layer protocol.

that governs the transfer of hypertext, which includes text, images, videos and other multimedia files, between web browsers and web servers. HTTP operates over the TCP (Transmission Control Protocol) and uses port 80 by default.

Key characteristics of HTTP include its stateless nature, client-server model, and the use of methods like GET and POST for requesting and submitting information. While HTTP is suitable for many web applications, it lacks encryption, making data transmission vulnerable to interception and tampering.

- HTTPS: HTTPS or Hypertext Protocol Secure, is the secure version of HTTP. It adds a layer of security to the communication between the client and the server by employing encryption. HTTPS uses SSL/TLS (Secure Sockets Layer/Transport Layer Security) protocols to encrypt the data, ensuring that it remains confidential and integral during transmission.

Key characteristics of HTTPS include Data Encryption, Authentication, Secure Transactions, Trust and Credibility.

HTTPS addresses this limitation by providing a secure and encrypted connection, making it an essential choice for websites that handle sensitive information or those aiming to establish trust and credibility with their users.

Nowadays, the industry trend is moving towards widespread adoption of HTTPS to ensure a safer and more private web browsing experience.

Section - C

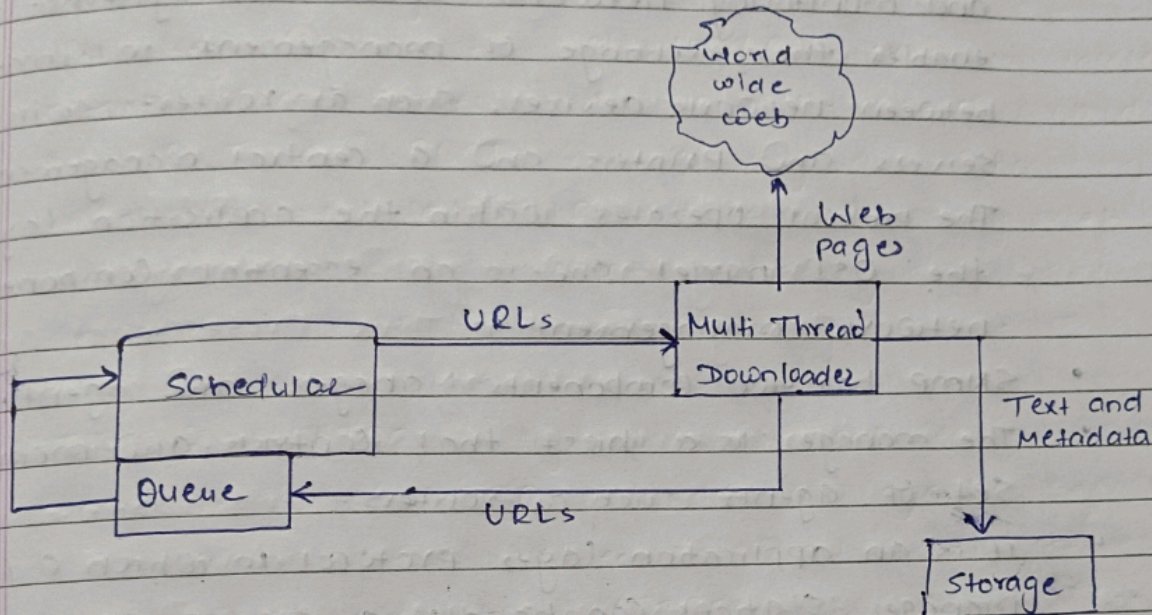
1. Explain World Wide Web with Architecture?

Ans The World Wide Web (WWW) is a of interlinked hypertext documents and multimedia content that is accessed via the internet using web browsers. The architecture of the World Wide Web involves several key components and standards that work together to enable the creation, publication and retrieval of information.

Here's an overview of the architecture of the World Wide Web:

- (i) Uniform Resource Identifier (URI):- The URI is a fundamental concept in the web architecture. It includes uniform Resource locaters (URLs) and Uniform Resource Names (URNs) that provide a standardized way to identify and locate resource on the web.
- (ii) Hypertext Transfer Protocol and Hypertext Transfer Protocol Secure (HTTPS):- HTTP is the protocol used for transferring hypertext (text, images, videos, etc.) on the web. It operates over the TCP/IP Protocol. HTTPS is the secure version of HTTP, employing SSL/TLS encryption to secure the communication between web browsers and servers.
- (iii) Web Browsers: Web browsers, such as Chrome, Firefox, Safari, and Edge, are user interface that allow individuals to access and interact with web content. Browsers interpret HTML (Hypertext Markup Language) document and render them as web pages.
- (iv) Web servers: Web servers store and serve web content in response to client requests. They use HTTP or HTTPS Protocols to communicate with

- (v) Web Development Frameworks and Libraries:- Developers often use frameworks and libraries, such as React, Angular and Vue.js, to streamline the process of building interactive and responsive web applications.



Here the browser displays a web page on the client machine when the user clicks on a line of text that is linked to a page on `abd.com`, the browser follows the hyperlink by sending a message to the `abd.com` server asking for the page.

• Features of WWW:

- HyperText information system
- Cross - Platform
- Distributed
- Open Standards and Open source
- "Web 2.0"

• Components of the web: There are 3 components of the web.

- (i) Uniform Resource Locator (URL): serves as a system for resource identification.
- (ii) HyperText Transfer Protocol (HTTP): specifies comm. of browser and server.
- (iii) HyperText Markup Language (HTML): defines the structure, organization and content of a web page.

Q 2. State ^{and} ~~the~~ explain simple network management Protocol.

Ans. Simple Network Management Protocol, commonly known as SNMP, is a widely used protocol for managing and monitoring network devices and systems. SNMP enables the exchange of management information between network devices, such as routers, switches, servers and printers, and a central management system. The protocol operates within the application layer of the OSI model and is an essential component in network management.

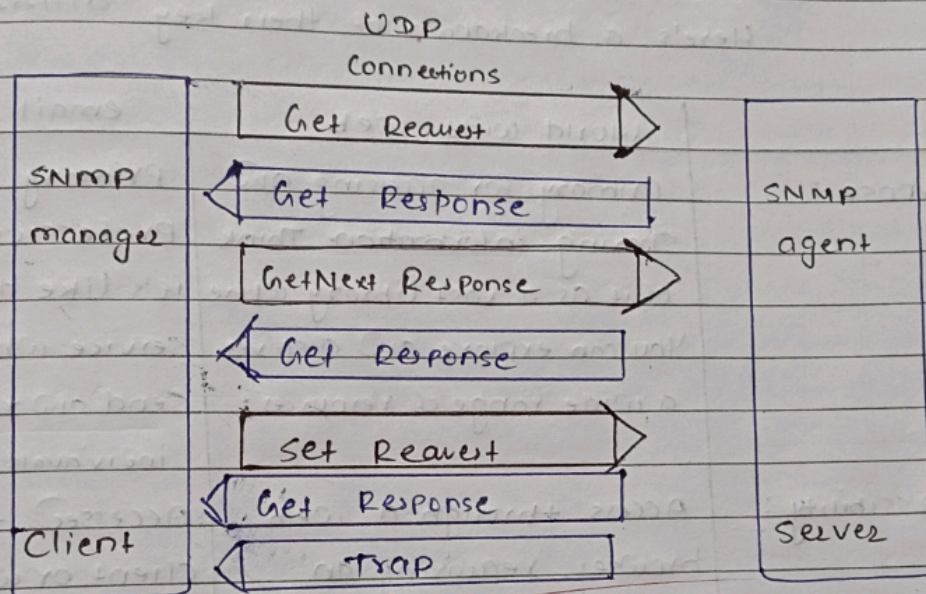
- SNMP has two components Manager and agent.
- The manager is a host that controls and monitors a set of agents such as routers.
- It is an application layer protocol in which a few manager stations can handle a set of agents.

Key Components of SNMP:

- (i) Manager: The SNMP manager is a system or application responsible for monitoring and managing network devices. It collects and analyzes information from SNMP agents.
- (ii) Agent: SNMP agents are software modules that run on managed network devices. These agents collect and store management information and respond to requests from SNMP managers.
- (iii) Managed Devices: These are the network devices being monitored and managed using SNMP. Examples include routers, switches, servers and various networked equipment.

(iv) **SNMP Messages:** SNMP uses a set of standardized messages to exchange information between the manager and agent.

SNMP defines five types of Message: GetRequest, GetNextRequest, SetRequest, GetResponse, and Trap.



SNMP Operations:

(i) **GET Request** :- The SNMP manager sends a GET Request to an SNMP agent, requesting the value of a specific parameter from the MIB.

(ii) **SET Request** :- The SNMP Manager sends a SET request to an SNMP agent, instructing it to modify the value of a particular parameter in the MIB.

(iii) **GETNEXT Request** :- The SNMP manager sends a GETNEXT request to retrieve the next value in the MIB, facilitating sequential retrieval of MIB data.

(iv) **GET BULK** :- this operation allows the manager to retrieve large amounts of data from the MIB in a single request.

(v) **TRAP** :- SNMP agents can send TRAP notifications to the SNMP manager to alert it about specific events or conditions such as critical errors or system reboots.

3. differentiate betⁿ. The world wide web and electronic mail.

Ans While both the world wide web and email allow for communication and access to information, they are distinct technologies with different purpose and functionalities.

Here's a breakdown of their key differences:

	World wide web	email
(i) Purpose:	Primary for accessing and sharing information. Think of it as a vast library where you can explore and discover a wide range of resources.	Primary for personal and professional communication. It's like a digital postal service where you can send and receive message with individuals or groups.
(ii) Accessibility:	Access through a web browser, requiring an internet connection.	Accessed through an email client or web interface, also requiring an internet connection.
(iii) Communication Model	One-to-many communication model.	One-to-one or one-to-many communication model.
(iv) Security	Security concerns vary depending on the website and protocol used. HTTPS provides encryption for secure communication.	Security can be vulnerable to phishing attacks and secure protocols like TLS can be used for encryption and authentication.
(v) Scalability	The web is highly scalable and can accommodate a vast amount of information and users.	Email scalability depends on the email provider's infrastructure.

(vi) Content and Structure:

Content is organized in hyperlinked web pages, allowing users to navigate easily. Pages can contain text, images, videos and interactive elements formatted using HTML and CSS.

Primarily text-based, with optional attachments like images and documents. Emails follow a specific format (with sender, recipient, subject line, and body text).

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